

Quantifying Competitive Excellence: A Bayesian Hierarchical Approach to Latent Skill Estimation in Strategic Game Shows Forest Evolutionism 2025

Emily Fan

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1 Introduction

1.1 Program Background

Forest Evolutionism is a strategic competition Chinese reality variety show produced by Mango TV. The program invites contestants with diverse industry backgrounds (Finance, Law, Actor, Game live streamer etc.) to participate in a series of complex board games. Each game is governed by unique and intricate rule sets that require players to make optimal decisions with their calculation, observation, negotiation, communication, collaboration and competition skills, making game results ranking an ideal subject for quantitative skill analysis.

1.2 Research Motivation and Objective

As a viewer of the program, I am interested in how the ranking of each game can reflect a player's ability. The first challenge in evaluating player's ability from raw game result is that the relative strength of opponents differs in matches, the second challenge is the inherent stochasticity and differing mechanics across different game setting. Motivated by the strategic depth of Forest Evolutionism, this project investigates the latent skill of each player by employing Bayesian data analysis. This approach quantifies uncertainty and provide a more robust and probabilistic ranking of the 22 contestants based on observed rankings.

2 Data Description

2.1 Data Source and Format

The raw data is manually derived from *Forest Evolutionism 2025* program series. Since no public API or structured dataset exists for this program, I personally curated a proprietary dataset by performing manual data extraction from 50+ hours of video broadcast on official @MangoTV-Mystery YouTube channel [4]. This involved identifying categorical game mechanics and mapping rank-order outcomes into a structured CSV format.

To ensure the reliability of the latent skill estimation and reduce noise from highly irregular game formats, a specific filtering logic was applied:

- **Game Selection:** Only games with at least two matches were included to allow for the estimation of consistent game-specific effects.
- **Player Participation:** The analysis focuses on games featuring individual or two-player team formats, where individual contribution to the final ranking is most discernable.

The final structured dataset comprises 92 player-match records across five distinct games (Forest, Invest, Ocean, Escape, and Nature).

2.2 Exploratory Data Analysis

In Figure 1, the heatmap shows players' rankings across different matches (darker blue indicates better performance), with the number of matches played and average ranking on the right. An inherent characteristic of this dataset is the unbalanced participation frequency, certain players competed in multiple game types while others appeared only in select games. This data sparsity justifies the use of a Bayesian framework to handle varying levels of uncertainty. The filtering criteria (minimum 2 matches per game type) resulted in a sparse matrix where some players have high visibility across multiple games while others appeared only in select matches. This deliberate selection emphasizes "quality of evidence" over "quantity of data," justifying the use of a Bayesian framework to handle the resulting unbalanced sample sizes.

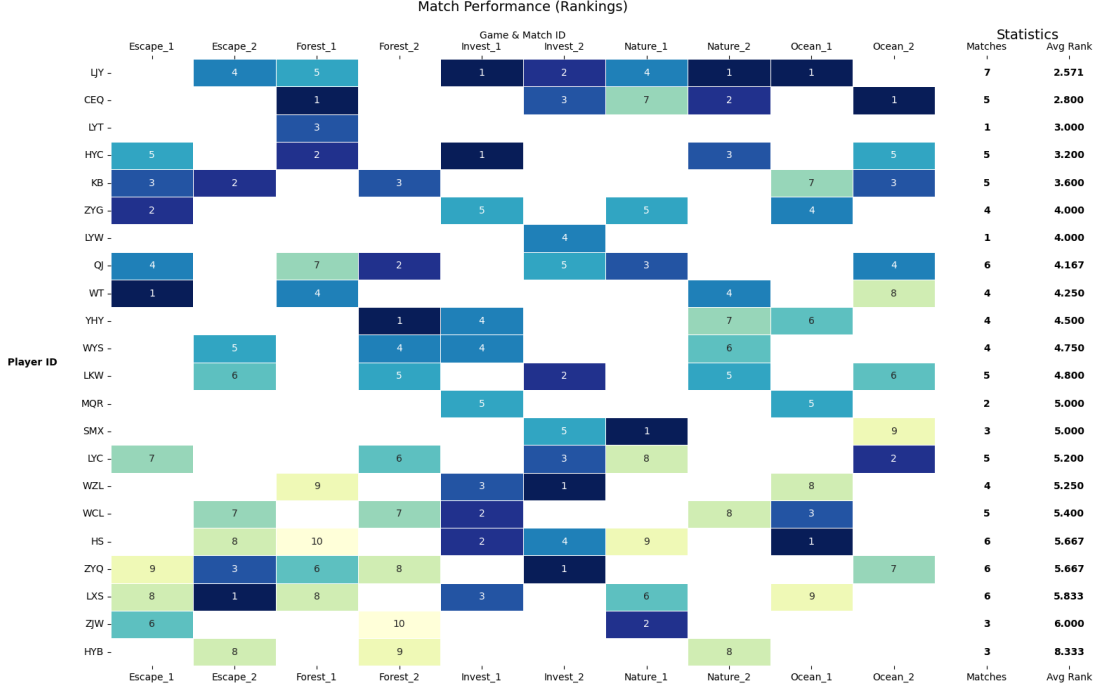


Figure 1: Player Participation and Performance Heatmap.

3 Methodology

3.1 Hierarchical Ranking Model

Instead of a simple win-loss comparison, a Hierarchical Linear Model is implemented to estimate player skills from rankings. The ranking of a player in a specific match is modeled as a continuous latent variable influenced by their inherent ability and the specific game environment.

The observed rank y_{ij} for player i in game j is modeled as:

$$y_{ij} \sim N(\mu_y, \epsilon^2)$$

$$\mu_y = -(\text{Skill}_i + \text{GameEffect}_j)$$

The negative sign ensures that a higher skill value corresponds to a lower numerical rank (e.g., Rank 1).

3.2 Prior Distributions

The model employs a Hierarchical Bayesian structure to handle data sparsity and provide shrinkage for players participating fewer matches.

- **Player Skill:** A non-centered parameterization to improve convergence:

$$\text{Skill}_i = \mu_p + \tau_i \cdot \sigma_p$$

$\tau_i \sim N(0, 1)$ is the raw individual effect, μ_p is the mean and σ_p is the standard deviation.

- **Game Effect:** Each game type j is assigned a random effect $\text{GameEffect}_j \sim \text{Normal}(0, 1)$ to ensure that player’s skill estimates are not biased by the specific mechanics of a single game.
- **Observation Noise:** $\epsilon \sim \text{Half-Normal}(1)$ is selected to constrain the variance to be positive while maintaining the assumption that noise in match rankings is relatively small.

3.3 DAG

Figure 2 shows the graphviz Digraph of the model.

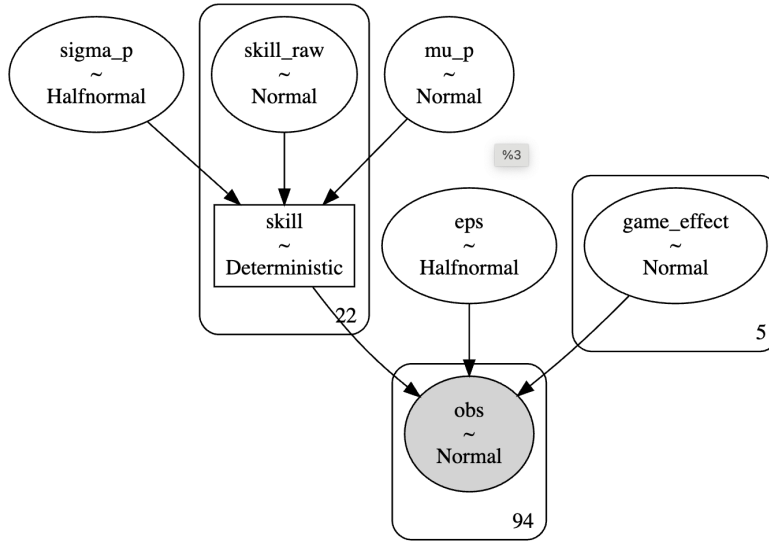


Figure 2: DAG graph of model

3.4 Computational Details

Ranks were zero-indexed for computational convenience. Inference was performed using PyMC with the No-U-Turn Sampler. To ensure high-quality posterior estimates, 4 chains with 10,000 samples each, following a 2,000-sample burn-in period were generated and a high target_accept=0.99 was set to mitigate potential divergent transitions. Model fit was evaluated using Posterior Predictive Checks via arviz.

4 Results and Discussion

4.1 Convergence Diagnostics

As shown in Figure 3, the overlap between the observed ranking distribution and the simulated datasets from the posterior indicates that the model accurately captures the statistical properties of the "Forest Evolutionism" game results. The MCMC chains were monitored for convergence using trace plots. Figure 4 displays a selected subset of trace plots for representative players. The posterior density plots (left) reveal a distinct unimodal posterior distribution for the latent skill parameters and the trace plots (right) show all four chains traversing the parameter space with excellent mixing, confirming that a stationary distribution was achieved.

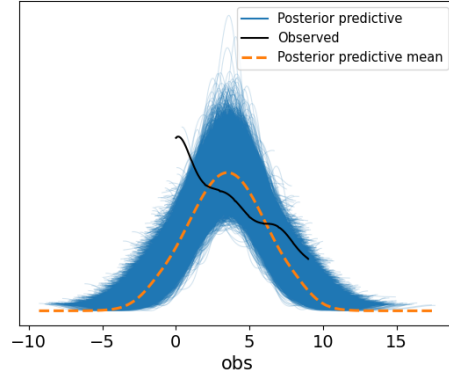


Figure 3: Posterior Predictive Check Plot

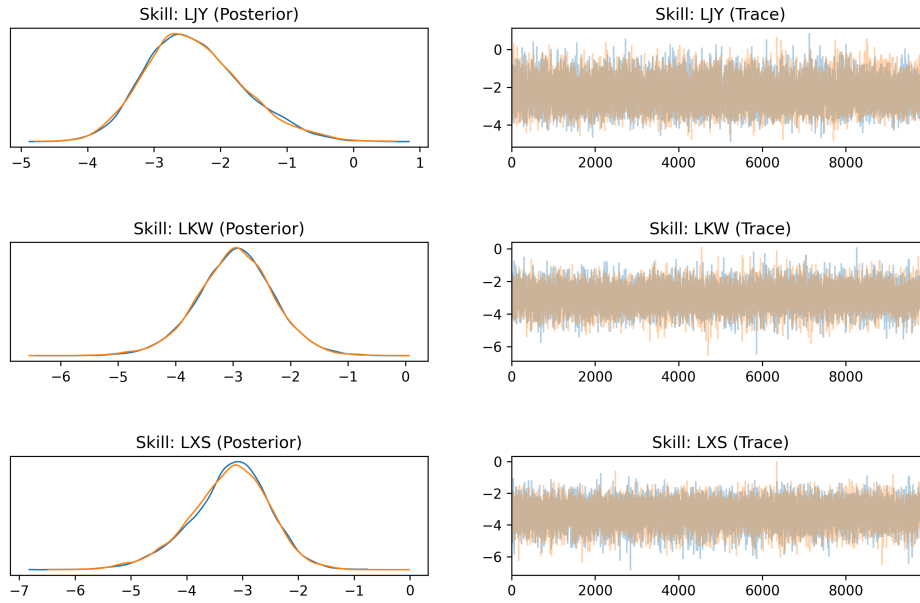


Figure 4: Selected MCMC Trace Plots for players.

4.2 Skill Estimates and Ranking

The main output of the hierarchical model is the posterior distribution of the latent skill parameter λ_i for each of the 22 players. Figure 5 illustrates the ranked posterior means along with their 94% Highest Density Intervals.

- **Top Performers:** Players at the top of the chart (e.g. LJY, CEQ, HYC) shows highest latent skills. These estimates accounts for both their ranking (skills) as well as the game effects they participated in.
- **Uncertainty and HDI:** The width of the HDI bars shows the model's uncertainty. Players who participated in more matches (e.g. LJY - 7 matches; QJ, HS, ZYQ, LXS - 6 matches) generally show narrower intervals, indicated a more precise estimates. For players who participated in less matches (e.g. LYT, LYW - 1 match; MQR - 2 matches; ZJW, HYB - 3 matches), a much wider intervals is observed, indicating model's more caution in the estimate.

- **Shrinkage Effect:** Individual estimates are slightly shrunk toward the population mean μ_p . This regularization prevents overestimation based on a single "lucky" win or an "unlucky" loss.

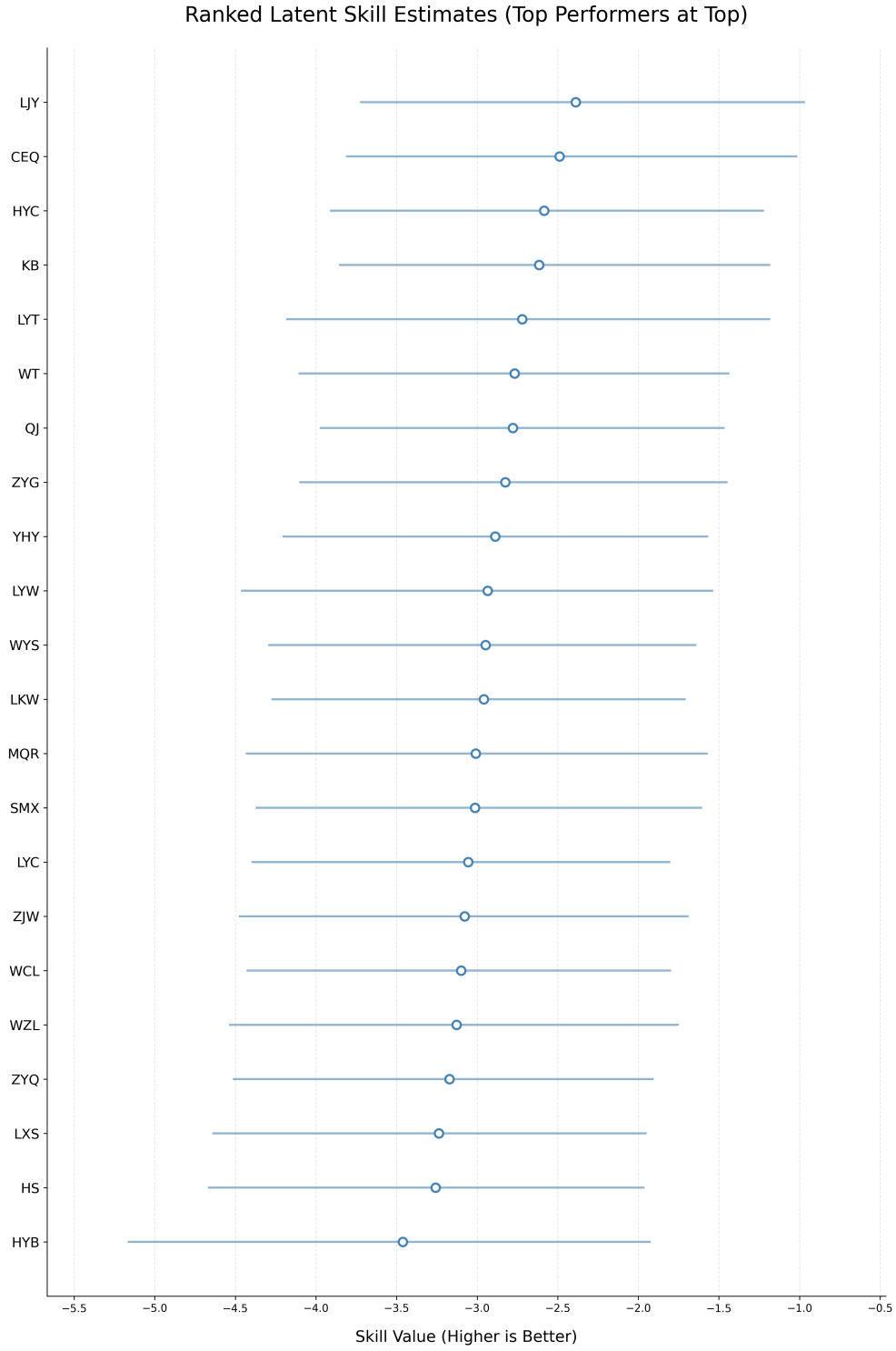


Figure 5: Ranked Latent Skill Estimates with 94% Highest Density Intervals (HDI).

4.3 Result Summary

Set of trace plots for 20 players is included in Appendix B. The summary table 1 shows summary result for top 3 and last 2 players, the full summary is included in Appendix C.

- **Convergence Confirmation:** $\hat{R}$ is exactly 1.0 for all players, and the Effective Sample Size exceeds 9,000 for all contestants.
- **Precision and Uncertainty:** The standard deviation remains relatively consistent across players (ranging from 0.67 to 0.88). The slight variations in SD and HDI widths reflect the differing amounts of match data available for each player, proving the model’s ability to quantify uncertainty for sparse records.
- **Skill Differentiation:** The top-ranked player, LJY (Mean = -2.388), shows a significant latent skill advantage over the lowest-ranked player HYB (Mean = -3.461). The overlapping HDIs shows that while the mean rankings provide a point estimate of skill, the relative performance between many mid-tier players contains inherent uncertainty.

Player	Mean	SD	3% HDI	97% HDI	MCSE	ESS	R-hat
LJY	-2.388	0.739	-3.726	-0.965	0.008	9537	1.0
CEQ	-2.488	0.748	-3.814	-1.014	0.007	10065	1.0
HYC	-2.585	0.714	-3.912	-1.221	0.007	10763	1.0
...	...	...	...	...	...	...	...
HS	-3.258	0.716	-4.670	-1.961	0.007	10573	1.0
HYB	-3.461	0.876	-5.168	-1.923	0.009	9253	1.0

Table 1: MCMC results for the skill parameter of top 3 and last 2 players

4.4 Winning Probability Analysis

Winning probabilities is also computed with the Bayesian model. By comparing the latent skill samples across all 10,000 iterations, the likelihood of each player attaining the top rank in a hypothetical match is estimated.

As shown in Table 2, LJY holds the highest winning probability at 16.72% nearly double that of the 4th-ranked player, followed by CEQ (12.76%) and HYC (9.20%). While the skill mean differences between top players appear small, the cumulative effect in a competitive setting results in clear probabilistic setting. The sharp decline in probabilities from the top tier to the bottom tier (where HYB stands at 0.87%) highlights the model’s ability to differentiate skill levels effectively.

Player ID	Win Prob.	Player ID	Win Prob.
LJY	16.72%	MQR	2.93%
CEQ	12.76%	LKW	2.63%
HYC	9.20%	SMX	2.61%
KB	8.20%	ZJW	2.28%
LYT	7.70%	LYC	2.07%
WT	5.30%	WZL	1.92%
QJ	4.41%	WCL	1.64%
ZYG	4.14%	ZYQ	1.44%
LYW	4.10%	LXS	1.22%
YHY	3.60%	HS	1.12%
WYS	3.17%	HYB	0.87%

Table 2: Winning Probabilities for All Contestants (Ranked)

5 Conclusion

This project successfully developed and implemented a Bayesian Hierarchical Model to evaluate the latent skills of 22 players in selected matches in Forest Evolutionism 2025.

5.1 Key Findings

The model identified LJY as the top-performing player with a latent skill estimate (λ) of -2.388 and 16.72% winning probability followed by CEQ ($\lambda = -2.488, 12.76\%$) and HYC ($\lambda = -2.585, 9.20\%$). The analysis revealed a clear hierarchy among the players while quantifying the uncertainty associated with each estimate through 94% HDIs. The use of hierarchical structure allowed the model to effectively handle data sparsity, preventing overestimation due to small sample sizes.

5.2 Limitations and Future Work

While the model achieved outstanding convergence, limitations such as not accounting specific interaction between different players game style (e.g. aggressive vs. defensive, calculation vs. intuition) or the strategic suppression exhibited by other contestants, who may collaboratively target high-performing players to hinder their ability to achieve top rankings. Future work can work on exploring how these affect the data. Furthermore, the current model treats player and game effects as nominal categorical variables. A more advanced iteration could incorporate Hierarchical Partial Pooling to share information across players with sparse match records (e.g., those with only 1 match), or utilize an Ordered Probit link function to better reflect the ordinal nature of ranking data rather than assuming Gaussian-distributed noise on the latent scale.

References

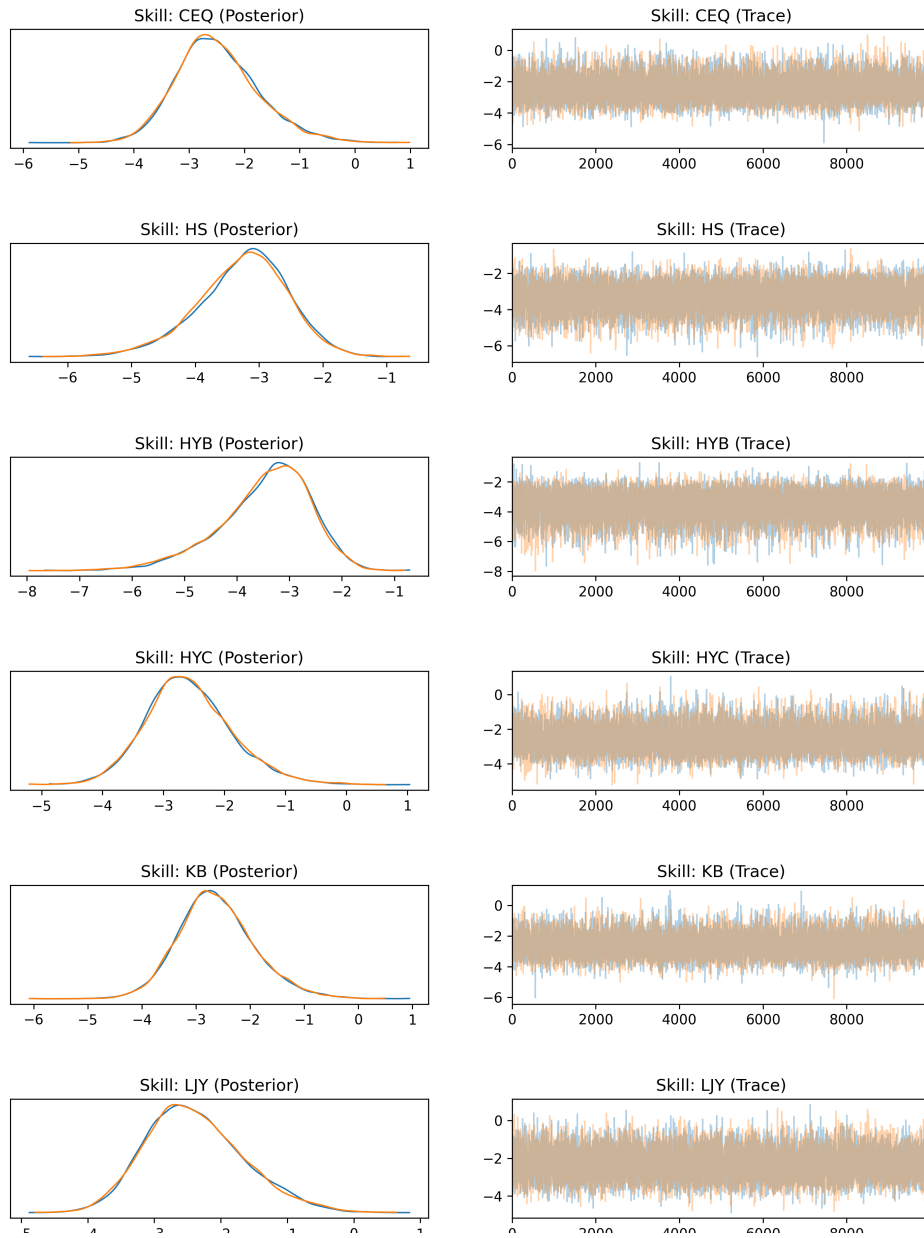
- [1] Vidakovic, B. (2017). *Engineering Biostatistics: An Introduction with Case Studies*. Wiley.
- [2] Abril-Pla, O., et al. (2023). PyMC: A modern, and comprehensive interface for Bayesian non-centered, hierarchical modeling. *PeerJ Computer Science*, 9, e1516.
- [3] Kumar, R., et al. (2019). ArviZ a unified library for exploratory analysis of Bayesian models in Python. *Journal of Open Source Software*, 4(33), 1143.
- [4] @MangoTV-Mystery. (2025). *Forest Evolutionism 2025 Full Episodes* [Video playlist]. YouTube. https://www.youtube.com/playlist?list=PL4nGKjc_PJOSLprB-EZiwa6UB6uBht5x5

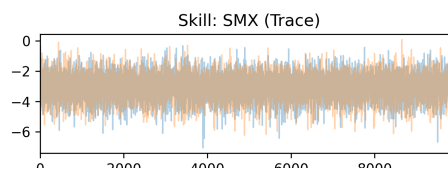
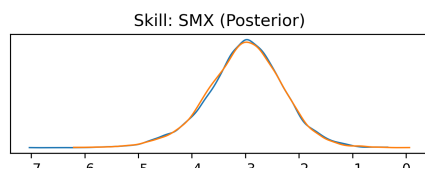
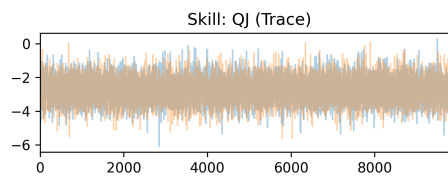
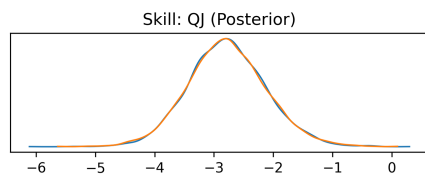
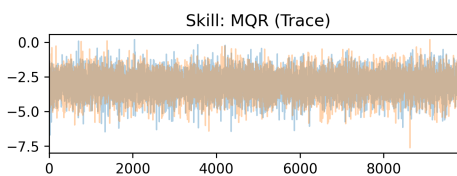
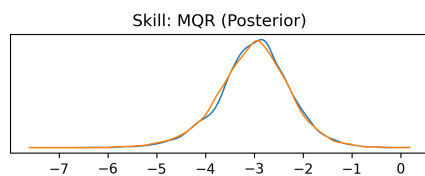
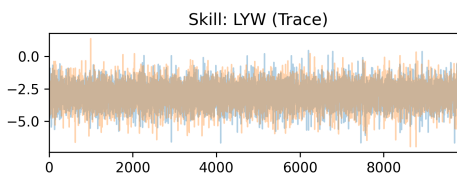
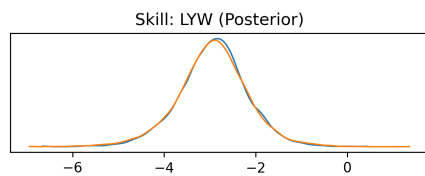
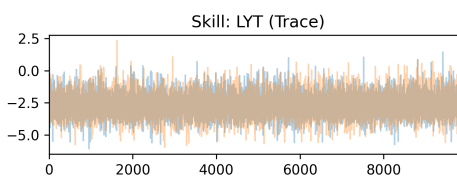
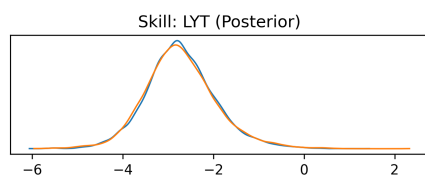
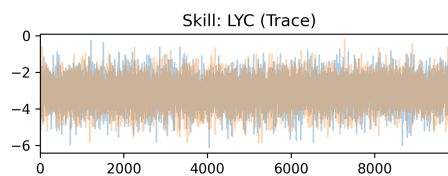
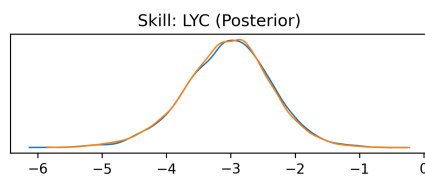
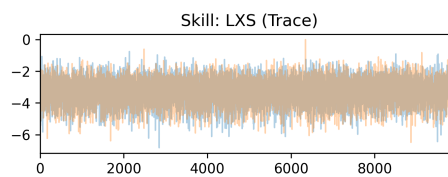
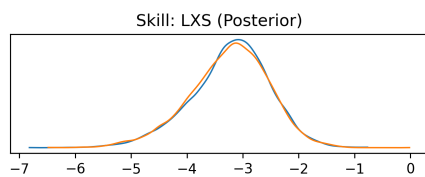
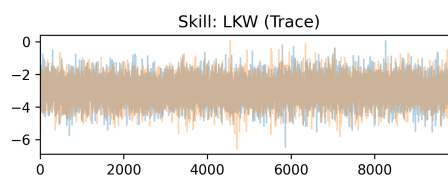
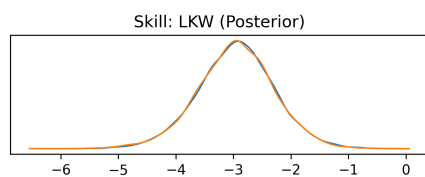
A Raw Match Data

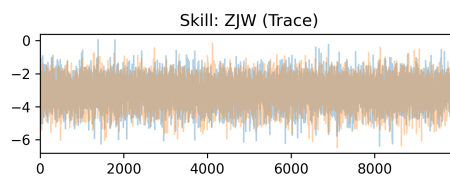
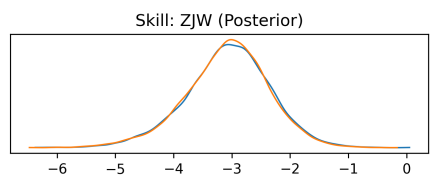
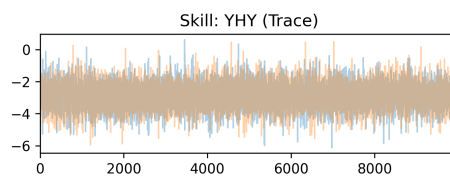
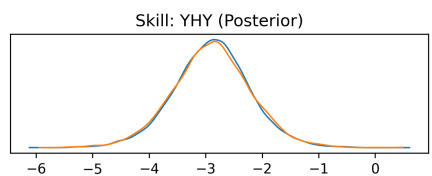
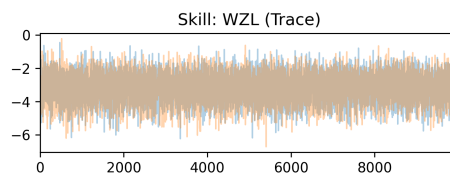
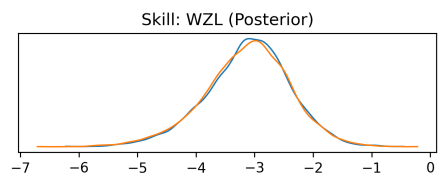
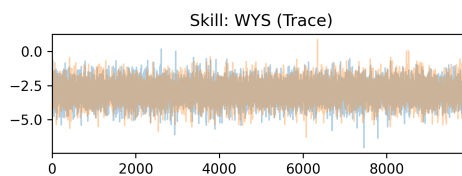
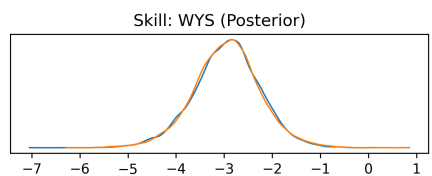
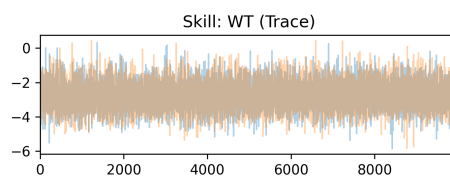
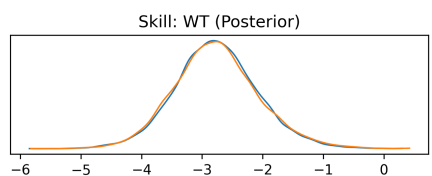
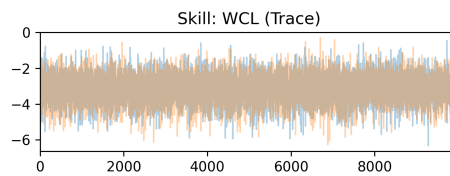
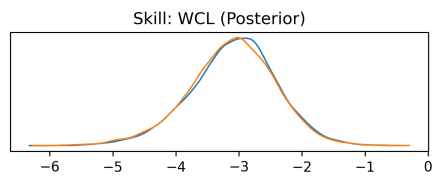
Below is the processed dataset derived from the match records.

Player ID	Game	Match	Rank		Player ID	Game	Match	Rank
CEQ	Forest	1	1		HS	Ocean	1	1
HYC	Forest	1	2		LJY	Ocean	1	1
LYT	Forest	1	3		WCL	Ocean	1	3
WT	Forest	1	4		ZYG	Ocean	1	4
LJY	Forest	1	5		MQR	Ocean	1	5
ZYQ	Forest	1	6		YHY	Ocean	1	6
QJ	Forest	1	7		KB	Ocean	1	7
LXS	Forest	1	8		WZL	Ocean	1	8
WZL	Forest	1	9		LXS	Ocean	1	9
HS	Forest	1	10		CEQ	Ocean	2	1
YHY	Forest	2	1		LYC	Ocean	2	2
QJ	Forest	2	2		KB	Ocean	2	3
KB	Forest	2	3		QJ	Ocean	2	4
WYS	Forest	2	4		HYC	Ocean	2	5
LKW	Forest	2	5		LKW	Ocean	2	6
LYC	Forest	2	6		ZYQ	Ocean	2	7
WCL	Forest	2	7		WT	Ocean	2	8
ZYQ	Forest	2	8		SMX	Ocean	2	9
HYB	Forest	2	9		WT	Escape	1	1
ZJW	Forest	2	10		ZYG	Escape	1	2
HYC	Invest	1	1		KB	Escape	1	3
LJY	Invest	1	1		QJ	Escape	1	4
HS	Invest	1	2		HYC	Escape	1	5
WCL	Invest	1	2		ZJW	Escape	1	6
WZL	Invest	1	3		LYC	Escape	1	7
LXS	Invest	1	3		LXS	Escape	1	8
YHY	Invest	1	4		ZYQ	Escape	1	9
WYS	Invest	1	4		LXS	Escape	2	1
MQR	Invest	1	5		KB	Escape	2	2
ZYG	Invest	1	5		ZYQ	Escape	2	3
WZL	Invest	2	1		LJY	Escape	2	4
ZYQ	Invest	2	1		WYS	Escape	2	5
LKW	Invest	2	2		LKW	Escape	2	6
LJY	Invest	2	2		WCL	Escape	2	7
LYC	Invest	2	3		HYB	Escape	2	8
CEQ	Invest	2	3		HS	Escape	2	8
HS	Invest	2	4		SMX	Nature	1	1
LYW	Invest	2	4		ZJW	Nature	1	2
QJ	Invest	2	5		QJ	Nature	1	3
SMX	Invest	2	5		LJY	Nature	1	4
ZYG	Nature	1	5		LYC	Nature	2	8
LXS	Nature	1	6		HS	Nature	1	9
CEQ	Nature	1	7		LJY	Nature	2	1
LYC	Nature	1	8		CEQ	Nature	2	2
HYC	Nature	2	3		WT	Nature	2	4
LKW	Nature	2	5		WYS	Nature	2	6
YHY	Nature	2	7		WCL	Nature	2	8
HYB	Nature	2	8					

B MCMC Trace Plots







C Complete Posterior Summary Statistics

Player	Mean	SD	3% HDI	97% HDI	MCSE	ESS	R-hat
LJY	-2.388	0.739	-3.726	-0.965	0.008	9537	1.0
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KB	-2.616	0.708	-3.857	-1.182	0.007	11166	1.0
LYT	-2.721	0.789	-4.184	-1.181	0.007	12821	1.0
WT	-2.768	0.703	-4.108	-1.435	0.006	12625	1.0
QJ	-2.778	0.667	-3.976	-1.465	0.006	12048	1.0
ZYG	-2.826	0.692	-4.103	-1.446	0.006	12525	1.0
YHY	-2.888	0.695	-4.206	-1.566	0.006	12382	1.0
LYW	-2.935	0.767	-4.464	-1.535	0.006	14512	1.0
WYS	-2.948	0.697	-4.295	-1.640	0.006	12325	1.0
LKW	-2.959	0.676	-4.276	-1.706	0.006	12119	1.0
MQR	-3.008	0.747	-4.435	-1.569	0.007	13082	1.0
SMX	-3.013	0.724	-4.374	-1.604	0.006	12753	1.0
ZJW	-3.078	0.731	-4.478	-1.689	0.006	13035	1.0
LYC	-3.055	0.686	-4.399	-1.801	0.006	11940	1.0
WCL	-3.100	0.695	-4.431	-1.796	0.006	11521	1.0
WZL	-3.128	0.730	-4.538	-1.749	0.007	11404	1.0
ZYQ	-3.172	0.693	-4.515	-1.904	0.007	10776	1.0
LXS	-3.237	0.713	-4.641	-1.949	0.007	10547	1.0
HS	-3.258	0.716	-4.670	-1.961	0.007	10573	1.0
HYB	-3.461	0.876	-5.168	-1.923	0.009	9253	1.0

Table 3: Full MCMC results for the skill parameter of all 22 contestants.